# Component Template

**Page Content**



## Overview

What is the purpose of this component; just a short text.

### Concepts

(Only if component has new concepts that need to be understood in order to use it)

### Architecture

(Only if component is sufficiently complex and interacts with other parts of the system)

### Implementation

No details, just a few generic points how it works internally.

## Usage

This is how the component can be instantiated in the system.

### Declaration of Component in CMake

Some short text about how to declare the component in the CMakeLists.txt, wether it needs a XXX_client as well and what them params of the DeclareCAmkESComponent_XXX() are:

**<Component> Usage 1**

```
DeclareCAmkESComponent_XXX(
        <NameOfComponent>
)
```

### Configuration of Component in system_config.h

Some short text about the #defines to be set here and what they mean:

**<Component> Usage 2**

```
#define XXX_PARAM_1 10
#define XXX_PARAM_2 11
```

## Instantiation and Configuration in CAmkES

Some short text about CAmkES setup:

### Declaring the Component

**<Component> Usage 3**

```
#include "XXX/XXX.camkes"
DECLARE_COMPONENT_XXX(<ComponentTypeName>, ...)
```

Component should provide this macro, even if there are no parameters. This leave the user the freedom to define one or more type name and don't enforce anything.

### Connecting a Component

**<Component> Usage 4**

```
#include "XXX/XXX.camkes"
component <ComponentTypeName> <ComponentInstanceName>
CONNECT_INSTANCE_XXX(<ComponentInstanceName>, ... )
```

Convenience macro to connect the standard interface of a component instance.

### Instantiating and Connecting the Component

**<Component> Usage 5**

```
#include "XXX/XXX.camkes"
DECLARE_AND_CONNECT_INSTANCE_XXX(<ComponentTypeName>, <ComponentInstanceName>, ...)
```

Convenience macro to create a component instance and connect the default interfaces. The recommendation is not to provide this macro unless there is a real benefit here. The CAmkES code is usually easier to understand if component instances are created explicitly and the macro **`CONNECT_INSTANCE_XXX()`** described above is used.

### Configuring the Instance

**<Component> Usage 6**

```
CONFIGURE_INSTANCE_XXX(<NameOfInstance>, ...)
```

# Example

Describe example for the usage for all the relevant parts.

## Instantiation of Component in CMake

See above, just with concrete names and parameters.

## Configuration of Component in system_config.h

See above, just with concrete names and parameters.

## Instantiation and Configuration in CAmkES

See above, just with concrete names and parameters.

## Using the Component's Interfaces in C

Maybe add some minimal code examples, e.g. highlighting how to use client-side libraries. But keep it minimal!